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COMMENT

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Comment

Comment on 'Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics'

Namit Anand

National Institute of Science Education and Research, Bhubaneswar, India

E-mail: namit.anand@niser.ac.in

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1. Proof

In [1], section 2.3 (*Superposition rules in the formalism of quantum theory*), the authors define a superoperator G[1] and go on to prove that any superposition of eigenstates of G with different eigenvalues is not physically admissible. The eigenstates of operator G are of the form $|g_m;\ \alpha_m\rangle$ where g_m and α_m are labels used to distinguish different eigenvectors of G, all of which are non-degenerate, i.e. $\langle g_m;\ \alpha_m|g_n;\ \alpha_n\rangle = 0$ if $m \neq n$.

In section 2.4 (*Impossibility of superposing states belonging to different coherent sectors*), they construct a vector $|u\rangle = \sum_m u_m|g_m;\ \alpha_m\rangle$ superposing the eigenstates of G and claim that since G commutes with every other observable of the system, $|u\rangle$ should be an eigenstate of every such operator in a complete set of commuting observables of the system.

Now we know that linear combination(s) of non-degenerate eigenvectors is not an eigenvector of the same operator i.e. take a superposition of two eigenvectors of G, $|u\rangle = a|g_m;\ \alpha_m\rangle + b|g_n;\ \alpha_n\rangle$ where $a \neq b$ in general, then

$$G|u\rangle = G(a|g_m;\ \alpha_m\rangle + b|g_n;\ \alpha_n\rangle) = a * g_m|g_m;\ \alpha_m\rangle + b * g_n|g_n;\ \alpha_n\rangle. \tag{1}$$

Note that this equation is not an eigenvalue equation for G, since $G|u\rangle \neq \lambda|u\rangle$ unless $a = b$, which is not true in general or if $g_m = g_n$, which is not true from the fact that the eigenstates of G are non-degenerate.

It is true that if G commutes with every other observable and has non-degenerate eigenvalues, then all such observables have the same eigenvectors. But $|u\rangle$ itself is not an eigenvector of G as proved above. And so $|u\rangle$ cannot be an eigenvector of other observables too. This is the erroneous assumption made by the authors. To summarize, the proof of

[1] A superoperator is one which commutes with all observables of the system.

limitations on the superposition principle is thus not valid for the case as discussed by the authors. Whether a similar proof can be constructed is still open.

Reference

[1] Cisneros C, Martínez-y-Romero R P, Núñez-Yépez H N and Salas-Brito A L 1998 Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics *Eur. J. Phys.* **19** 237